

Getting started with Amazon SageMaker

AWS Prescriptive Guidance

Amazon SageMaker is a fully managed service that provides every developer and data scientist with the ability to build, train, and deploy machine learning (ML) models quickly. This guide covers the ML lifecycle on SageMaker, including data preparation with Data Wrangler, training at scale, experiment tracking, model deployment with real-time and batch inference, MLOps with Pipelines, feature engineering with Feature Store, and best practices for production ML workloads.

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Table of Contents

- [Introduction](#)
- [SageMaker Studio](#)
- [Data Preparation](#)
- [Model Training](#)
- [Experiment Tracking](#)
- [Model Deployment and Inference](#)
- [SageMaker Feature Store](#)
- [SageMaker Pipelines \(MLOps\)](#)
- [Security and Governance](#)
- [Cost Optimization](#)
- [Best Practices](#)
- [FAQ](#)

Introduction

Amazon SageMaker accelerates every stage of the machine learning lifecycle, from data preparation and model training to deployment and monitoring. SageMaker removes the heavy lifting from each step of the ML process so teams can develop high-quality models more quickly. SageMaker provides tools for experienced ML engineers, collaborative notebooks for data scientists, and AutoML capabilities for developers and analysts new to machine learning.

SageMaker supports all major ML frameworks including TensorFlow, PyTorch, Apache MXNet, scikit-learn, XGBoost, and Hugging Face Transformers. It provides optimized containers for each framework with pre-installed libraries and dependencies, and allows you to bring your own containers for maximum flexibility.

SageMaker Studio

Amazon SageMaker Studio is a fully integrated development environment (IDE) for machine learning. It provides a unified web-based interface where you can perform every step of ML development, including preparing data, writing code, training models, deploying models, and monitoring performance — all from a single pane of glass. Studio is built on JupyterLab and supports multiple kernels, Git integration, and collaboration features.

SageMaker Studio Notebooks are persistent, on-demand notebooks that run on dedicated compute instances. Unlike classic SageMaker notebooks, Studio notebooks are fully managed in the cloud and support fast startup with pre-warmed instances. Studio also includes integrated tools for experiment tracking, model registry, feature store, pipelines, and monitoring.

Data Preparation

SageMaker Data Wrangler

SageMaker Data Wrangler is the fastest and easiest way to prepare data for ML. Data Wrangler includes over 300 built-in data transformations, so you can quickly normalize, transform, and encode data. It provides a visual interface to explore and understand data, identify data quality issues, and apply transformations without writing code. Data Wrangler integrates with Amazon S3, Athena, Redshift, SageMaker Feature Store, and third-party data sources.

SageMaker Processing

SageMaker Processing enables you to run preprocessing, postprocessing, and model evaluation workloads at scale. You can use built-in containers for scikit-learn, Spark, or bring your own container. Processing jobs are fully managed, automatically provisioning and tearing down compute resources,

allowing you to focus on writing transformation logic rather than managing infrastructure.

Model Training

SageMaker Training provides managed, scalable infrastructure for training ML models. You specify the training algorithm (built-in, framework, or custom container), the training data location in S3, the instance type and count, and hyperparameters. SageMaker automatically provisions the compute cluster, trains the model, and saves the model artifacts to S3 when training is complete.

Training Type	Use Case	Key Feature	Managed Infrastructure
Single-instance training	Small to medium datasets	Simple, fast setup	Yes
Distributed data parallel	Large models, large datasets	SageMaker Distributed	Yes
Distributed model parallel	Very large models (LLMs)	Tensor/pipeline parallelism	Yes
Spot training	Cost-sensitive training	Up to 90% cost reduction	Yes
Hyperparameter Tuning (HPO)	Automated hyperparameter search	Bayesian optimization	Yes
SageMaker Autopilot	Automated ML (AutoML)	Full ML pipeline automation	Yes

SageMaker HyperPod

SageMaker HyperPod is a purpose-built infrastructure for training and fine-tuning large foundation models. HyperPod provides Slurm- and Kubernetes-orchestrated resilient clusters for large-scale training. It includes automatic node health monitoring and replacement, checkpointing to resume from the last saved state when a node fails, and persistent storage with FSx for Lustre. HyperPod supports GPU clusters with EFA networking for extremely low-latency, high-bandwidth inter-node communication.

Experiment Tracking

SageMaker Experiments enables you to track, compare, and evaluate ML experiments. An experiment is a collection of runs. Each run captures the code version, input parameters, hyperparameters, metrics, and output artifacts. SageMaker Studio provides a visual interface for comparing experiment runs across dozens of metrics simultaneously. Experiments integrate with SageMaker Autopilot, Training Jobs, and Pipelines.

Model Deployment and Inference

SageMaker provides multiple deployment options to match your latency, throughput, and cost requirements.

Deployment Type	Latency	Use Case	Billing
Real-Time Inference	Milliseconds	User-facing APIs, real-time predictions	Per second, endpoint is running
Serverless Inference	< 1 second cold start	Infrequent, spiky traffic	Per inference + duration
Async Inference	Seconds to minutes	Large payloads, long processing	Per inference + duration
Batch Transform	Minutes to hours	Offline scoring, large datasets	Per instance hour
Multi-Model Endpoints (MME)	Milliseconds	Many models, same framework	Per instance hour, shared

SageMaker Feature Store

SageMaker Feature Store is a fully managed, purpose-built repository to store, share, and manage features for machine learning. Feature Store consists of an online store (low-latency, for real-time inference) and an offline store (Amazon S3-backed, for training). Features can be written once and served from both stores, eliminating the training-serving skew problem where features computed at training time differ from features computed at inference time.

SageMaker Pipelines (MLOps)

SageMaker Pipelines is a purpose-built, easy-to-use continuous integration and continuous delivery (CI/CD) service for machine learning. With Pipelines, you can create, automate, and manage end-to-end ML workflows that include data preparation, model training, model evaluation, and model registration and deployment. Pipelines integrates with the SageMaker Model Registry, enabling automated model approval and deployment workflows.

Best Practices

- Use Managed Spot Training for cost savings: Spot Training can reduce training costs by up to 90% for fault-tolerant training jobs. Implement checkpointing to recover from interruptions.

- Use SageMaker Feature Store to eliminate training-serving skew: Ensure that features computed for training and inference use the same transformation logic.
- Implement experiment tracking from the start: Use SageMaker Experiments to track every training run, making it easy to reproduce results and compare model versions.
- Register all models in the Model Registry: Use the SageMaker Model Registry as the single source of truth for approved model versions and their metadata.
- Use multi-model endpoints for cost efficiency: When serving many models of the same framework type, group them behind a Multi-Model Endpoint to reduce hosting costs.
- Monitor models in production: Use SageMaker Model Monitor to detect data drift, model quality degradation, and bias in deployed models.
- Right-size your training and hosting instances: Use instance recommendations from SageMaker Inference Recommender to identify the best instance type for cost and latency requirements.
- Implement CI/CD for ML with Pipelines: Automate training, evaluation, and deployment pipelines to enforce quality gates before deploying new model versions to production.

Frequently Asked Questions

Q: What is the difference between SageMaker Autopilot and SageMaker Canvas?

A: SageMaker Autopilot is an AutoML service targeted at data scientists who want full transparency into the generated notebooks and code. SageMaker Canvas is a no-code ML interface for business analysts to build and use ML models without writing any code.

Q: Can I use my own Docker containers with SageMaker?

A: Yes. SageMaker supports Bring Your Own Container (BYOC) for training, processing, and inference. You can extend AWS-provided base images or build from scratch, as long as your container implements the SageMaker training or serving interface.